



DATA PROVIDERS AGREE ON REPORTED NUMBERS, AND NOT ON ESTIMATES

RESEARCH · PRIVATE MARKETS · CLIMATE DATA · AI

Climate risk in private markets: the data gap, and what AI can and can't fix

Private capital is growing fastest where climate data is thinnest. We read the evidence on how climate risk reaches private portfolios, how AI now fills the missing numbers, how far off those numbers can be, and what GPs and LPs should do about it.

US\$16–16.5tn

in private-markets funds in 2025, plus US\$8–8.5tn in other private capital¹

as much as 90%

of private-fund emissions data that LPs have to estimate¹⁶

2.4×

less effective: estimated data at spotting the worst emitters¹³

10 Nov 2026

first measured emissions due from large private US companies¹⁷

In one minute

- **The money is private; the data is not.** Private-markets funds held US\$16–16.5tn¹ in 2025, and 87%²⁰ of US firms with more than US\$100m revenue are private. Most of them publish no emissions.
- **So most numbers are guesses.** LPs say as much as 90%¹⁶ of the emissions data they get from private funds is estimated.
- **The risk is real and already priced.** About half of 2025's disaster losses were uninsured, flood-exposed homes sell at a discount, and insurers are leaving exposed areas.
- **AI helps, within limits.** Machine-learning estimates beat simple averages, but they mostly learn a company's size and sector. They were 2.4×¹³ less effective at finding the worst emitters, and no study has tested them on private companies.
- **Regulators allow estimates, if you say so.** What gets punished is calling an estimate a measurement, or overselling the AI behind it.
- **The fix is a ladder, not a model.** Estimate now, label every number with a quality score, and replace estimates with measured data on a set timetable.

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1. How big private markets are

"Private markets" means investments that do not trade on a stock exchange. That covers private equity, private credit, infrastructure and real estate. Estimates of their size depend on the definition, so we quote each source with its own.

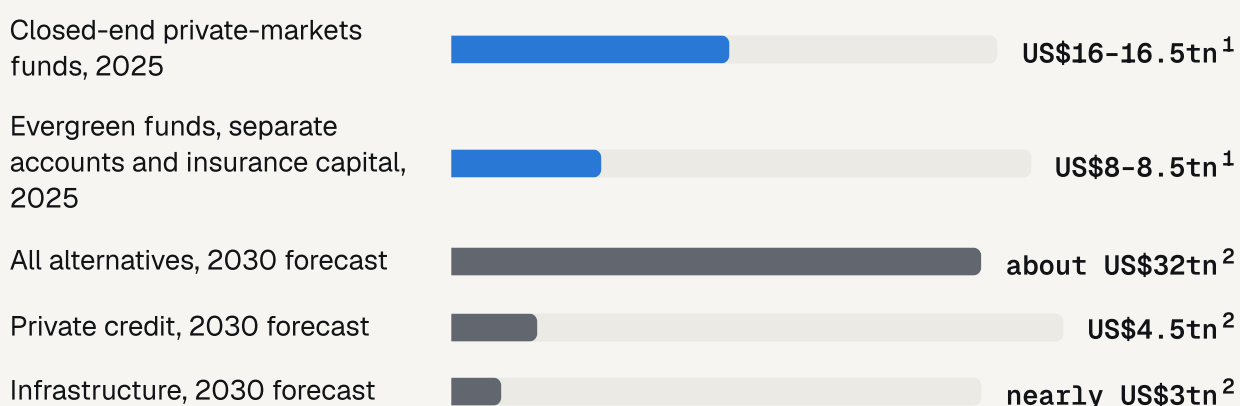
The most recent full-year figures are McKinsey's, published in February 2026 and covering 2025. Traditional closed-end private-markets funds held US\$16–16.5tn¹. On top of that sat another US\$8–8.5tn¹ in other forms of private capital: evergreen funds, separately managed accounts and insurance balance sheets. Private equity deal value reached US\$2.6tn¹ in 2025.

Looking ahead, Preqin forecasts that all "alternatives", which also include hedge funds and natural resources, will reach about US\$32tn² by 2030. It expects private credit to reach US\$4.5tn² and infrastructure nearly US\$3tn².

Private capital today, and where it is heading

US\$ trillion. 2025 figures are McKinsey's, drawn at the midpoint of each range. 2030 figures are Preqin's forecasts, which also count hedge funds and natural resources. The two use different definitions, so do not add them together.

■ 2025, McKinsey ■ 2030 forecast, Preqin



A large share of the real economy is also private. 87%²⁰ of US companies with more than US\$100m in revenue are privately owned. Meanwhile the number of US listed companies fell from more than 8,000 in 1996 to 4,572²¹ by early 2023.

IN PLAIN ENGLISH

Stock-market rules are how most climate data gets published. A growing share of large companies are not on a stock market, so those rules never reach them.

2. The data gap, in numbers

The clearest single number comes from the investors in private funds. Members of ILPA, the association of limited partners, say that as much as 90%¹⁶ of the private-fund emissions data they receive has to be estimated, not reported by the company.

The industry's shared dataset

The ESG Data Convergence Initiative (EDCI) is the main effort to fix this. Its members agree to collect the same metrics from their portfolio companies, with Scope 1 and 2 emissions mandatory and Scope 3 optional.⁴⁰

500+ GPs and LPs in the ESG Data Convergence Initiative ⁵	9,000 portfolio companies in the latest data cycle ¹⁸	230,000+ data points collected ¹⁸
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The industry's shared dataset keeps growing

Total assets managed by firms that belong to the ESG Data Convergence Initiative, US\$ trillion. This is the members' whole balance sheet, not the private assets the data covers.

January 2024 · 375+ members		~US\$28tn ³
July 2025 · 450+ members		>US\$38tn ⁴
2026 · 500+ members		~US\$59tn ⁵

Take care with EDCI's headline numbers. The trillions are the *whole* balance sheets of the member firms, mostly large LPs and asset managers. They are not the value of private assets that the data covers. Membership grew from 375+ in January 2024³ to 450+ in July 2025⁴ and 500+ in 2026.⁵

Joining is not the same as having a plan. Using EDCI data for 2023, BCG found that only a minority of PE-owned companies had a climate strategy or a net-zero target.

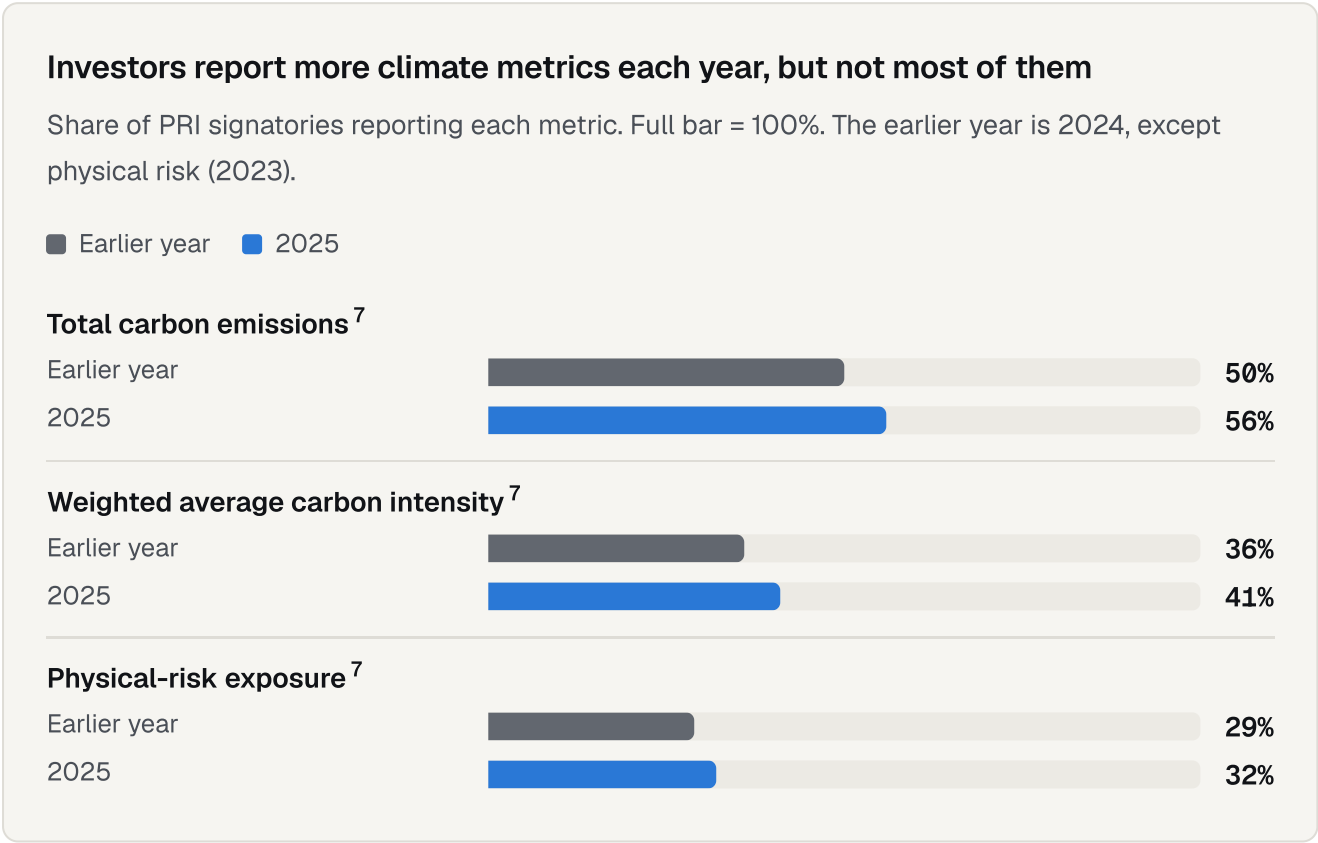
Few PE-owned companies have a climate plan

Share of companies in the EDCI dataset, 2023 data. Full bar = 100%.

Have a decarbonisation strategy		22% ⁶
Committed to net zero by 2050		17% ⁶
Median share of renewable electricity		30% ⁶

EDCI's benchmark of estimated against actual data is shared only with members. **No public figure shows what share of EDCI companies report measured emissions, or how fast that share is rising.** That is the single most useful number this market does not publish.

Investors themselves report only part of the picture



Outside private markets, 22,000+²² companies disclosed through CDP in 2025, nearly 11,000²² of them through its simpler questionnaire for small and medium-sized firms. CDP does not split them into private and listed companies.

Scoring the quality of each number

PCAF, the Partnership for Carbon Accounting Financials, gives every financed-emissions figure a data-quality score from 1 (best) to 5 (worst). Its standard, now in a third edition covering ten²³ asset classes, is what most banks and investors use.

	Score 1	Emissions the company reported, checked by an auditor.
	Score 2	Emissions the company reported, not checked.
	Score 3	Estimated from what the company physically uses: energy bills, fuel, production.
	Score 4	Estimated from the company's revenue and its sector's average.
	Score 5	Estimated from the sector average and the size of the investment alone.

Scores 1–3 follow PCAF's hierarchy.²³ The wording of scores 4 and 5 follows earlier editions and has not yet been checked against the 2025 text.

Most private-markets holdings sit at scores 4–5 in early inventories. The one public asset-owner example we found is the Finnish pension insurer Varma. Its portfolio-weighted score was 2.96²⁴, with asset classes ranging from 1.32 to 5.0²⁴. No one publishes average PCAF scores by private asset class.

IN PLAIN ENGLISH

A score of 4 or 5 means "we multiplied the company's revenue by an industry average". It tells you how big the company is, not how clean it is.

3. Why it matters: the risk is already showing up

Physical risk: losses, insurance and prices

About half of 2025's disaster losses were not insured

Natural-catastrophe losses in 2025, US\$ billion, as estimated by three firms. Each uses its own method; do not average them. Swiss Re's economic total was not available.

■ Economic losses ■ Insured losses

Munich Re⁸



Aon⁹



Swiss Re Institute¹⁰



The three estimates differ, but they tell the same story. Aon puts the uninsured share of 2025 losses at 51%⁹, the lowest on record. That low figure is flattered by the US, which accounted for 81%⁹ of insured losses; elsewhere far less is insured. Swiss Re finds that 92%¹⁰ of insured losses came from "secondary perils": frequent, local events such as wildfire, storms and floods. Those are exactly the hazards that hit individual buildings, plants and farms.

In the US, NOAA counted 27²⁵ billion-dollar disasters in 2024, costing US\$182.7bn²⁵. NOAA then retired that database in 2025, so there is no official 2025 figure.⁴¹

about 7%

discount on US homes exposed to sea-level rise. Still about 4%²⁶ for homes not expected to flood for about a century.

48 of the 50

US counties with the highest insurance non-renewal rates are flood- or wildfire-exposed, according to a Senate committee analysis the industry disputed.⁴²

29.4%

projected rise in home insurance premiums by 2055, wiping US\$1.47tn²⁷ off US property values. This is one vendor's model, not an observation.

about 16%

higher sale prices for green-certified US offices, with rents about 3%²⁸ higher.

IN PLAIN ENGLISH

Climate risk reaches an asset's value through its insurance bill and its buyer before any flood arrives. Real estate and infrastructure feel it first because they cannot move.

Transition risk: the cost of carbon is rising

29%

of global emissions now carry a direct carbon price¹⁹

87

carbon taxes and trading schemes in force¹⁹

~US\$21/t

average price per tonne of CO₂e¹⁹

>US\$107bn

raised by carbon pricing in 2025¹⁹

Carbon pricing is spreading, but it is still partial and cheap on average.¹⁹ The central banks' own climate scenarios became more severe in their latest update: NGFS Phase V raised its estimate of 2050 losses from chronic physical risk by 2 to 4 times²⁹. Stress tests built on older scenarios probably understate the downside. For fossil fuels, one study puts upstream oil and gas assets at risk of stranding at more than US\$1tn³⁰, with most of the loss falling on private investors in rich countries, including pension funds.

High-carbon assets are moving into private hands

When listed companies sell polluting plants to meet climate goals, the pollution does not fall; it changes owner. A 2025 study in the *Journal of Finance* found emissions unchanged after the sale, while the seller's ESG rating improved.⁴³ Two advocacy groups have put numbers on the flow. The Environmental Defense Fund counted 155 deals worth US\$86.4bn³¹ between 2017 and 2021 that moved oil and gas assets away from net-zero-committed companies. The Private Equity Stakeholder Project found that 65%³² of the energy companies owned by the 20 largest PE firms were fossil-fuel companies, emitting about 1.17bn tCO₂e³² a year. Neither figure has been independently verified.

IN PLAIN ENGLISH

A private buyer can pick up a brown asset cheaply. It also picks up the stranding risk, and the LP questions that come with it.

4. The rules that demand the numbers

By September 2026 the rules had split in two. Rules for *managers and asset owners* still require portfolio-wide emissions. Rules for *companies* have mostly been narrowed, except in California.

RULE	WHAT IT ASKS OF PRIVATE-MARKETS INVESTORS	STATUS, SEP 2026
EU SFDR	Report emissions of investee companies (Scope 1–3) as principal adverse impacts. Where data is missing, "reasonable assumptions" are allowed. ⁴⁴	IN FORCE
EU SFDR 2.0	Three product categories replace Articles 8 and 9. Entity-level impact reporting kept only for firms in CSRD scope. ⁴⁵	PROPOSED Council position 24 Jun 2026 ⁴⁶ ; Parliament committee vote 10 Sep 2026 ⁴⁷
EU CSRD after Omnibus	Only companies with more than 1,000 staff and €450m turnover. Smaller firms can refuse value-chain data requests beyond the voluntary SME standard. ⁴⁸ About 80% of companies leave scope. ⁴⁹	IN FORCE 18 Mar 2026
ISSB IFRS S2	Financed emissions (Scope 3 Category 15) for asset managers, with some categories now excludable. ⁵⁰ Adopted by 28 ³³ jurisdictions.	FINAL amendments effective 1 Jan 2027
Australia AASB S2	Climate statements including financed emissions. Group 2 captures asset owners with at least AUD 5bn. ⁵¹	IN FORCE Group 2 from 1 Jul 2026
UK FCA ESG sourcebook	Entity and product climate reports with emissions and carbon intensity, for managers with at least £5bn, due 30 June each year. ⁵²	IN FORCE

RULE	WHAT IT ASKS OF PRIVATE-MARKETS INVESTORS	STATUS, SEP 2026
US SEC climate rule	Listed registrants only; never applied to private funds.	STAYED rescission proposed 29 May 2026 ⁵³
California SB 253	Scope 1, 2 and later 3 for US companies, public or private , with more than US\$1bn revenue doing business in California. ⁵⁴	ENFORCEABLE first reports due 10 Nov 2026 ¹⁷
California SB 261	Biennial climate-risk reports.	BLOCKED by court order since 18 Nov 2025 ⁵⁵

IN PLAIN ENGLISH

If you manage EU, UK or large Australian money, you must report your portfolio's emissions whether or not your companies measure them. That is why estimates exist. California's SB 253 is the one rule that makes large private companies measure their own.

5. How AI fills the gaps

"AI-estimated emissions" covers four quite different techniques. Each solves a different problem.

1 · Spend-based and input-output models

Multiply a company's revenue or spending by an average emissions factor for its sector. This is what most data vendors actually use: Trucost's model has about 500³⁴ sectors. It is simple and covers everyone, but two companies in the same sector with the same revenue always get the same footprint.

Best for: a first, complete picture. PCAF score 4-5.

2 · Machine-learning estimates

Train a model on the thousands of listed companies that do report, using revenue, assets, staff, sector and location, then predict the rest. It beats a plain sector average, but the inputs are still mostly size and sector.

Best for: ranking a large portfolio. Still an estimate, still score 4-5.

3 · Language models that read documents

Pull reported figures out of annual reports, sustainability reports and supplier documents. A benchmark of 139³⁵ reports now exists to test them. They find numbers that exist; they cannot create numbers that don't.

Best for: collecting what companies already disclose. Can reach score 1-2.

4 · Satellites and location data

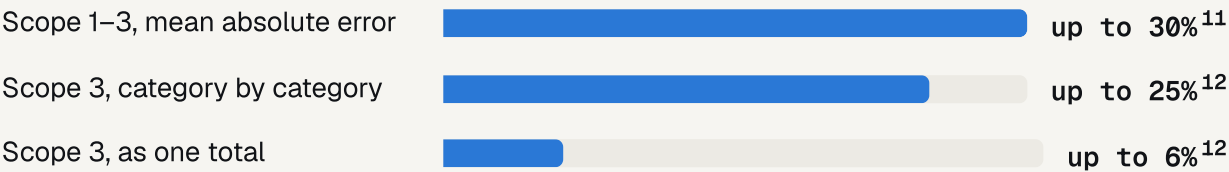
Map physical assets and measure or model their emissions and hazards directly. Climate TRACE tracks 2,765,771³⁶ emission sources built from 744,678,997³⁶ assets. Real estate and infrastructure have known coordinates, which is a genuine advantage for private markets.

Best for: real assets and physical risk.

How much better is machine learning?

Machine learning helps most where it matters least

Best reported improvement over earlier estimation models. Scope 3 — the supply chain, usually the biggest share — gains least.



The gains are real but relative: they are measured against older models, not against the truth. The authors of the Scope 3 study say plainly that "absolute prediction performance is low even with the best models".¹² The Scope 1-3 result is from a 2021 study using an ensemble of models.¹¹

IN PLAIN ENGLISH

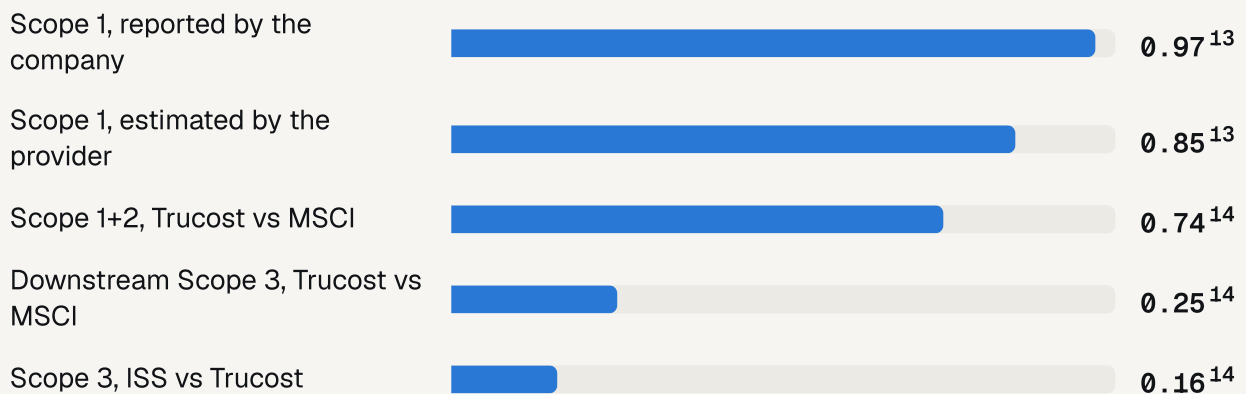
An AI estimate is a better-educated guess. It still mostly knows how big a company is and what it does, not how it runs.

6. How wrong can the estimates be?

The best test of an estimate is whether independent providers reach the same answer. When companies report, providers agree almost perfectly. When providers estimate, they drift apart, and for Scope 3 they barely agree at all.

Data providers agree on reported numbers, and not on estimates

Correlation between two providers' figures for the same companies. 1.0 = identical ranking, 0 = no relationship. Amundi reports its correlations as percentages (74%, 25%, 16%).

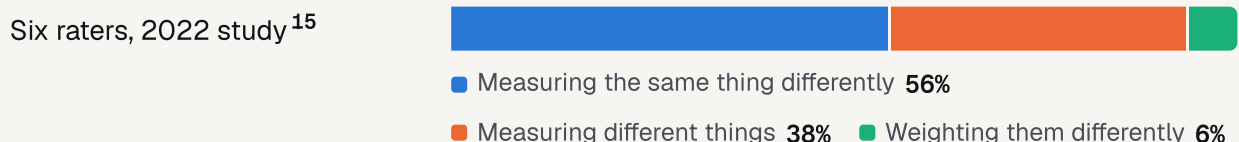


The most important finding for investors comes from the same study as the first two bars. Vendor estimates made up more than half of all emissions observations, and they were at least 2.4× less effective than reported data at identifying the worst emitters.¹³ For anyone trying to find the carbon hot-spots in a portfolio, that is the number that matters.

ESG ratings show the same pattern more broadly. A study of six raters found that most of their disagreement comes from measurement, not from different opinions about what matters.

Why ESG ratings disagree

Share of the disagreement between six ESG raters explained by each cause.



Physical-risk data is no more settled. GARP compared 13³⁷ physical-risk vendors and found they largely disagreed on which hazard mattered most for the same buildings.

Three limits to keep in mind

- **No test on private companies.** We found no study that trained or tested an emissions model on private-company data. Every model learns from listed companies and is then applied to a different population.
- **An estimate can't show progress.** If the number is revenue times a factor, a company that cuts emissions looks the same, and a year-on-year "reduction" may just be a revenue dip or a new factor.
- **Scope 3 is the weakest link.** It is usually the largest share of a footprint and the least predictable.

7. What regulators expect

No regulator we found bans estimates, and none has written rules specifically for AI-estimated emissions. They all ask for the same two things: **say what is estimated, and be able to back up what you claim.**

- **ESMA (EU)** asked for more transparency on data methods and on *the use of estimates* in its 2024 greenwashing report.⁵⁶
- **The EU ESG Ratings Regulation** has applied since 2 July 2026. Rating providers must apply to ESMA for authorisation by 2 November 2026.⁵⁷
- **IOSCO's** 2026 toolkit for supervising AI asks regulators to check for "hallucination": plausible but made-up outputs.⁵⁸
- **The US SEC** fined two advisers, Delphia US\$225,000³⁸ and Global Predictions US\$175,000³⁸, for claiming AI they did not use. Earlier ESG cases turned on the gap between stated and actual process, including a US\$19m³⁹ penalty for DWS.

IN PLAIN ENGLISH

Two separate risks attach to AI-estimated data. Calling an estimate a measurement is greenwashing. Overselling the model behind it is "AI-washing".

8. A five-step plan for GPs

The frameworks below all assume estimates are temporary. Put together, they form a ladder from estimates to measured data. The order is ours; no single body prescribes it.

1 Collect on one template.

Use the EDCI metric set, with Scope 1 and 2 mandatory. LPs already recognise it, and its guidance is updated each year.⁴⁰

2 Calculate the same way everywhere.

Apply the iCI greenhouse-gas standard for private equity, which applies the GHG Protocol and PCAF to portfolio companies and funds.⁵⁹

3 Score every number.

Put a PCAF data-quality score on each holding and publish the portfolio average, as Varma does.²⁴ It shows exactly how much is estimated, which is the transparency ESMA asks for.

4 Set a date for each estimate to become a measurement.

The IIGCC framework for private equity expects a net-zero ambition within one year of deal close and basic emissions management within two.⁶⁰ Deal close is the cheapest moment to install meters and data collection. Companies filing under SB 253 will produce measured Scope 1 and 2 from November 2026 anyway, so collect those figures.

5 For real assets, work asset by asset.

Use location-based physical-risk analysis rather than company-level scores. Run more than one vendor on material assets, feed the results into valuations and insurance plans,⁶¹ and stress-test with NGFS Phase V scenarios.²⁹

Using AI safely

- ✓ Label every figure as reported or estimated, with its PCAF score.
- ✓ Describe the model as what it is: "sector and revenue-based estimate", not "AI-measured".
- ✓ Keep a trail from each figure back to its inputs, so extraction errors and hallucinations can be caught.
- ✓ Where a company later reports, compare it with your old estimate and publish how far off you were.

9. Questions LPs should ask

Use these alongside the ILPA ESG Assessment Framework, updated in 2024 to reference ISSB and EDCI,⁶² and the PRI/ILPA/iCI Climate Module for due diligence.⁶³

1. What share of the fund's emissions is at PCAF score 4 or 5?
2. Which model produced the estimates, and has it ever been tested on companies like these?
3. Is the year-on-year trend driven by operations, or by revenue and factor changes?
4. By when will each estimate be replaced with measured data?
5. Are high-carbon assets bought from listed sellers covered by a transition plan?
6. For real assets: which physical-risk vendors were used, and did they agree?

10. What to watch in the next 12 months

- **Oct 2026** European Parliament plenary vote on SFDR 2.0, then negotiations with the Council. The outcome decides which impact indicators survive at product level.
- **2 Nov 2026** Deadline for ESG rating providers to apply for ESMA authorisation.
- **10 Nov 2026** First SB 253 filings: the first large sample of measured emissions from private US companies. It is also the first real chance to test AI estimates against private-company actuals.
- **1 Jan 2027** IFRS S2 amendments take effect, and relief under AASB S2025-1 begins in Australia.
- **2027** SB 253 Scope 3 reporting begins, and EU member states must transpose the Omnibus by March.

OUR VIEW

AI's lasting value in private markets is not in guessing missing company emissions. It is in two narrower jobs: locating physical risk for real assets, and extracting the data companies already hold. Treat estimates as scaffolding, and take them down on a published schedule. The first GP or vendor to publish a back-test of its estimates against SB 253 filings will turn "AI-estimated" from a disclosure risk into a method others can check.

11. Common questions

▼ Can AI estimate the emissions of a private company?

Yes, but mostly from its revenue, size and sector. Machine-learning models beat simple sector averages by up to 30%¹¹ on overall error, and by much less on Scope 3 as a total. No published study has tested them on private companies, so label the output as an estimate and score it PCAF 4 or 5.

▼ How much private-markets emissions data is estimated today?

ILPA members report that as much as 90%¹⁶ of the emissions data they receive for private funds is estimated.

▼ Do regulators allow estimated emissions?

Yes. SFDR lets managers use reasonable assumptions where data is missing.⁴⁴ What regulators penalise is presenting an estimate as measured, or overstating what an AI model does.

▼ What should a GP do first?

Collect Scope 1 and 2 on the EDCI template and put a PCAF score on every figure. That alone tells you, and your LPs, how much of your footprint is guesswork.

▼ Which rule forces private companies to measure emissions?

California's SB 253, for US companies with more than US\$1bn revenue doing business in California. First reports are due 10 Nov 2026¹⁷.

Sources

About the figures. Every figure in this briefing is attributed to the publisher named in the numbered source list, with the date that publisher gave it. The figures were compiled from publishers' pages and published summaries as of 27 September 2026. Some underlying documents could not be read in full, so check a figure against its source before relying on it. Educational analysis, not investment or legal advice.

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